
CF12: COMPLETE FORMALISM - GRAVITY, BLACK HOLES, AND SATURATION GEOMETRY IN VDM

COMPLETE FORMALISM

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ABSTRACT

CF12 closes the gravity and black-hole branch of the Complete Formalism program at the derived-limit level. It does not introduce primitive spacetime, primitive gravitation, or primitive horizon thermodynamics. Instead it inherits the primitive bifurcation invariant and non-discharge law from CF000, the derived carrier from CF00, the $J \oplus M$ reversible/irreversible split from CF01, the boundary-concentration hierarchy law from CF03, the finite-speed telegraph cone from CF04, and the dark-sector cosmological background from CF11, and then derives gravity as the carrier-level geometric response to invariant-burden gradients. The completed formalism consists of: (i) an equivalence principle in which inertial and gravitational mass are two measurements of the same burden functional; (ii) a Lorentzian metric whose null cone is fixed by the inherited transport cone; (iii) an Einstein-form effective field equation with an explicit saturation correction sector; (iv) Schwarzschild-form exterior geometry and horizon formation as same-domain saturation rather than annihilation; (v) black-hole entropy as boundary articulation count with the area law and Hawking temperature emerging from saturation geometry; (vi) information preservation and Page-curve behavior as articulation-budget transfer; and (vii) an explicit gravitational back-reaction correction term for the CF11 cosmogenesis residual. In this way CF12 is not a roadmap to gravity. It is the completed gravity module of the present VDM canon, with its low-energy closure, black-hole thermodynamics, and observational gates written as one coherent derived formalism.

Keywords gravity · black holes · saturation geometry · Hawking radiation · Page curve · derived Lorentzian metric · Einstein-form closure · VDM

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1 1. Executive Summary

CF12 is the gravity completion of the present Complete Formalism series. The paper is not allowed to assume a metric, a signature, a field equation, a horizon law, or a thermodynamic sector because those would all be stage leaks. The document therefore begins from the already-earned canon: CF000 provides non-discharge, saturation, and orthogonal re-articulation; CF00 provides the derived carrier $\mathcal{M}$ and smooth local variation; CF01 provides the induced $J \oplus M$ split together with invariant and entropy generators; CF03 provides hierarchical saturation and boundary concentration; CF04 provides finite-speed telegraph-Fisher cone structure; and CF11 provides the dark-sector cosmological background.

The central claim of CF12 is simple: gravity is the carrier-level geometry of nonuniform invariant burden. Once invariant-bearing articulation is concentrated nonuniformly on the carrier, the cost of lawful propagation changes from point to point, and the geometry seen by transport must deform accordingly. The unique local second-order divergence-compatible closure of that deformation law in four dimensions is Einstein-form. Horizons then appear not as places where articulation ends, but as radii where the same-domain articulation budget of a channel saturates. Once saturation occurs, CF000 forbids discharge, so the invariant must either remain borne on the saturation boundary or re-articulate orthogonally. Black-hole thermodynamics is the thermodynamic reading of that forced boundary storage and re-articulation process.

The completed formalism delivered here is therefore the following chain:

1. define invariant burden density and its carrier-level stress accounting;
2. derive equality of inertial and gravitational mass from one burden functional;
3. derive one distinguished propagation-limiting direction and the Lorentzian sign split from the CF04 cone;
4. write the effective gravitational field equation as an Einstein-form closure of burden-gradient geometry with explicit saturation corrections;
5. solve the static spherical vacuum sector and identify the horizon as same-domain saturation;
6. derive the area law, the Hawking temperature, and the evaporation ledger;
7. derive information preservation and a Page-curve law from articulation-budget transfer; and
8. close the CF11 cosmogenesis residual with an explicit gravitational back-reaction correction term.

2 2. Inherited Canon and Local Definitions

2.1 2.1 Inherited theorem ledger

CF12 uses the prior canon in the following exact way.

- **CF000.** The primitive bifurcation invariant does not discharge. Same-domain continuation saturates under redundancy. When saturation occurs, lawful continuation proceeds by orthogonal re-articulation rather than annihilation.
- **CF00.** The carrier domain $\mathcal{M}$ and smooth local variation are derived rather than assumed.
- **CF01.** The induced geometry splits into reversible and irreversible sectors $J \oplus M$, generated by an invariant-bearing functional I and an entropy-bearing functional Σ .
- **CF03.** Hierarchical interfaces concentrate articulation burden at boundaries under saturation; boundary concentration is therefore a theorem-bearing effect, not an imported holographic slogan.
- **CF04.** Propagation is finite-speed. The inherited transport cone is therefore real and load-bearing.
- **CF11.** The cosmological background already carries a dark-sector split. CF12 is allowed to add gravitational back-reaction to that background, not replace it.

2.2 2.2 Pre-geometric local burden

Definition 2.1 (Invariant burden density). Let $\rho_{\text{inv}}(x)$ denote the local density of invariant-bearing articulation at carrier point $x \in \mathcal{M}$. For a compact carrier region $\Omega \subset \mathcal{M}$, the total burden is

$$\mathcal{B}[\Omega] := \int_{\Omega} \rho_{\text{inv}} dV_g.$$

The quantity $\mathcal{B}[\Omega]$ is not a new primitive conserved charge. It is the carrier-level measure of how much invariant-bearing articulation must be hosted in Ω . The metric volume element dV_g is used here only as the carrier integration measure already inherited from CF00; the Lorentzian signature and its cone-normalized transport fields are not introduced until Section 4.

3. Equivalence Principle from Single-Burden Identity

The first gravity theorem cannot be the metric. Before there is geometry there must already be a reason that “inertial response” and “gravitational response” are numerically equal. In the present canon that reason is simple: both are measurements of the same burden functional.

Definition 3.1 (Inertial and gravitational mass functionals). For a compact burden host $\Omega \subset \mathcal{M}$, define

$$m_I(\Omega) := c^{-2}\mathcal{R}_J[\Omega], \quad m_G(\Omega) := c^{-2}\mathcal{R}_M[\Omega]. \quad (1)$$

Here $\mathcal{R}_J[\Omega]$ is the reversible resistance of the host to forced re-articulation of its transport state, while $\mathcal{R}_M[\Omega]$ is the coupling weight of the same host to an ambient burden gradient.

Theorem 3.2 (Equivalence principle from invariant identity). *For every compact carrier host Ω supporting invariant-bearing articulation,*

$$m_I(\Omega) = m_G(\Omega).$$

Proof. The reversible limb J measures how hard it is to deflect an already-borne articulation flow from its lawful transport channel. That resistance scales with the total burden that must be re-articulated, so the reversible response functional is $\mathcal{R}_J[\Omega] = \mathcal{B}[\Omega]$. The irreversible limb M measures how the same host couples to a nonuniform burden-density environment. A gradient can only couple to the burden actually borne by the host, so with the same carrier normalization the coupling functional is $\mathcal{R}_M[\Omega] = \mathcal{B}[\Omega]$. Because CF000 forbids replacing the burden by two independent primitives, there is no room for an independent inertial and gravitational substance behind those two operational readings. Hence

$$m_I(\Omega) = c^{-2}\mathcal{B}[\Omega] = m_G(\Omega).$$

Therefore their equality is derived from single-burden identity, not assumed by definition. $\square$

Corollary 3.3 (Universality of free fall). *All compact articulation hosts with the same initial carrier velocity experience the same gravitational acceleration in a given background burden gradient.*

Proof. The acceleration generated by a burden gradient is proportional to m_G/m_I . By the theorem, that ratio is identically one. $\square$

4. The Causal Cone and the Lorentzian Metric

4.1 One distinguished propagation-limiting direction

CF04 does not merely say that propagation is finite-speed. It says there is one finite-speed envelope that simultaneously bounds all propagation directions. That fact alone already distinguishes one direction from the rest.

Proposition 4.1 (One timelike direction from a common propagation bound). *Let the local transport law admit a finite-speed cone with limiting speed c_* acting simultaneously on all spatial propagation directions. Then there exists exactly one local ordering direction τ whose variation is propagation-limiting, while all other local directions are propagation directions relative to it.*

Proof. If two or more independent directions were fundamentally propagation-limiting, the carrier would have multiple distinct common clocks, and the cone would split into incompatible speed limits. CF04 gives one common limiting envelope, not several. Therefore one direction plays the role of common ordering/limiting direction, while the remaining directions are those whose propagation is bounded relative to it. $\square$

4.2 Lorentzian sign pattern

Theorem 4.2 (Lorentzian metric from the transport cone). *The minimal quadratic encoding of the inherited finite-speed cone has Lorentzian sign pattern. In local coordinates (τ, x^i) ,*

$$ds^2 = -c_*^2 d\tau^2 + h_{ij}(x) dx^i dx^j, \quad (2)$$

with h_{ij} positive definite on the propagation directions.

Proof. A Euclidean-signature quadratic form has no nontrivial null cone: $ds^2 = 0$ implies vanishing displacement in every direction. But CF04 delivers a genuine finite-speed boundary separating admissible from inadmissible propagation. The minimal quadratic encoding must therefore admit nontrivial null displacements. By the previous proposition, one direction is the universal ordering/limiting direction and the others are propagation directions. The required null condition is thus

$$-c_*^2 d\tau^2 + h_{ij} dx^i dx^j = 0,$$

which demands opposite sign class for the single limiting direction and the propagation directions. This is exactly Lorentzian signature. $\square$

Corollary 4.3 (Null cone/eikonal form). *For a phase function S whose level sets track carrier propagation fronts, the local eikonal relation is*

$$g^{\mu\nu} \partial_\mu S \partial_\nu S = 0. \quad (3)$$

Proof. Equation (3) is the dual statement that the phase gradient is null with respect to the metric whose null cone coincides with the inherited transport cone. $\square$

4.3 Metric-dependent carrier objects

Definition 4.4 (Carrier four-velocity and projector). Let u^μ be the future-directed unit carrier velocity field with respect to the Lorentzian cone metric derived in Theorem 4.2, normalized by

$$g_{\mu\nu} u^\mu u^\nu = -c_*^2.$$

Let

$$P_{\mu\nu} := g_{\mu\nu} + c_*^{-2} u_\mu u_\nu$$

be the spatial projector orthogonal to u^μ .

Definition 4.5 (Invariant source tensor). The effective source tensor accounting invariant burden is

$$T_{\mu\nu}^{\text{inv}} = \rho_{\text{inv}} u_\mu u_\nu + p_{\text{inv}} P_{\mu\nu} + \Pi_{\mu\nu} + q_{(\mu} u_{\nu)}. \quad (4)$$

Here p_{inv} is the isotropic burden pressure, q_μ is the burden-flux covector orthogonal to u^μ , and $\Pi_{\mu\nu}$ is the trace-free anisotropic burden stress. Equation (4) is the minimal symmetric tensorial bookkeeping of storage, transport, and anisotropic loading needed by the gravity module.

Definition 4.6 (Saturation scalar and correction tensor). Let $s(x)$ be a dimensionless saturation scalar with $s = 0$ in unsaturated weak-field regions and $s \rightarrow 1$ at a fully formed saturation boundary. The saturation correction tensor is defined by

$$\Delta_{\mu\nu}^{\text{sat}} := \alpha (\nabla_\mu \nabla_\nu s - g_{\mu\nu} \square s) + \beta (\nabla_\mu s \nabla_\nu s - \tfrac{1}{2} g_{\mu\nu} (\nabla s)^2), \quad (5)$$

with constants α, β fixed by weak-field and near-horizon matching. This sector vanishes when s is constant and is therefore inactive away from saturation gradients.

5 Burden-Gradient Geometry and the Effective Field Equation

5.1 Geometric response to nonuniform burden

Theorem 5.1 (Metric deformation from burden gradients). *Spatial and temporal gradients of ρ_{inv} deform the effective Lorentzian metric. To leading local order the geometry responds according to*

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \Delta_{\mu\nu}^{\text{sat}} = \kappa_{\text{inv}} T_{\mu\nu}^{\text{inv}}, \quad (6)$$

where $\kappa_{\text{inv}} = 8\pi G_{\text{eff}}/c^4$ and $\Delta_{\mu\nu}^{\text{sat}}$ is given by (5).

Proof. The geometry must encode how propagation cost changes under nonuniform burden. The left-hand side must therefore be: (i) local; (ii) symmetric rank-two; (iii) built from the metric and its derivatives; and (iv) divergence-compatible so that carrier source accounting remains balanced. In four dimensions the minimal such local second-order operator is Einstein-form, $G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu}$. CF12 adds one more sector that the standard Einstein form does not possess, namely the saturation correction $\Delta_{\mu\nu}^{\text{sat}}$, because CF00 and CF03 require explicit boundary saturation effects that vanish in ordinary weak-field regions but become non-negligible near horizon formation. The right-hand side is the minimal symmetric tensorial source accounting of invariant burden, transport, and anisotropy already defined in (4). Therefore the local geometric response is (6). $\square$

Remark 5.2. Equation (6) is the completed formal field equation of CF12. It is not written as a conjectural target. The standard Einstein equation is recovered as the derived limit $s \equiv 0$ or constant, in which case $\Delta_{\mu\nu}^{\text{sat}} = 0$.

5.2 5.2 Weak-field limit

Take the quasistatic weak-field ansatz

$$g_{00} = -c^2 (1 + 2\Phi/c^2), \quad g_{ij} = \delta_{ij} (1 - 2\Phi/c^2), \quad (7)$$

with $|\Phi| \ll c^2$, negligible anisotropic stress, negligible saturation gradients, and $u^\mu \approx (c, 0, 0, 0)$. Then (6) reduces to

$$\nabla^2 \Phi = 4\pi G_{\text{eff}} \rho_{\text{inv}}. \quad (8)$$

This is the Newtonian closure of the gravity module. It fixes the interpretation of G_{eff} and guarantees that burden density acts as the source of attractive potential in the weak-field regime.

5.3 5.3 Radial saturation diagnostic

For a static spherical burden concentration of total mass-equivalent M_{inv} , define

$$\chi(r) := \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r}. \quad (9)$$

The quantity $\chi(r)$ is the same-domain saturation ratio of the exterior channel. The exterior channel remains unsaturated when $\chi < 1$, reaches criticality at $\chi = 1$, and becomes horizon-screened for $\chi > 1$.

6 6. Static Spherical Sector and the Horizon

6.1 6.1 Schwarzschild-form exterior solution

In vacuum outside a static spherical source, $T_{\mu\nu}^{\text{inv}} = 0$ and s is constant, so (6) reduces to

$$G_{\mu\nu} = 0$$

for the exterior region. Use the standard static spherical ansatz

$$ds^2 = -e^{2\Phi(r)} c^2 dt^2 + e^{2\Lambda(r)} dr^2 + r^2 d\Omega^2. \quad (10)$$

The vacuum equations imply

$$e^{-2\Lambda(r)} = 1 - \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r}, \quad (11)$$

while asymptotic flatness and weak-field matching fix

$$e^{2\Phi(r)} = 1 - \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r}. \quad (12)$$

Therefore the exterior line element is

$$ds^2 = - \left(1 - \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r} \right) c^2 dt^2 + \left(1 - \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r} \right)^{-1} dr^2 + r^2 d\Omega^2. \quad (13)$$

6.2 6.2 Horizon as same-domain saturation

Theorem 6.1 (Horizon radius). *The horizon of the static spherical sector occurs at*

$$r_H = \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2}, \quad (14)$$

which is exactly the radius where $\chi(r_H) = 1$.

Proof. Equation (13) becomes null in the tt -component and singular in the rr -component when $1 - 2G_{\text{eff}}M_{\text{inv}}/(c^2r) = 0$. By definition (9), this is precisely the same-domain saturation threshold. The geometry therefore agrees with the primitive law: the horizon is where the exterior articulation channel saturates. $\square$

Remark 6.2. The horizon is not an annihilation surface. CF000 blocks that interpretation. What fails at r_H is the continuation of one *channel* of articulation, not the invariant itself. The invariant remains borne on the boundary and may re-articulate into other channels.

7 7. Black-Hole Thermodynamics as Boundary Burden Counting

7.1 7.1 Area law

CF03 says that under saturation, burden concentration migrates to interfaces and boundaries. The horizon is the saturation boundary of the gravitational channel. The correct entropy count is therefore a boundary count.

Let the fundamental articulation cell area on a saturation boundary be ℓ_*^2 . Then the number of horizon boundary cells is

$$N_H = \frac{A_H}{\ell_*^2}, \quad A_H = 4\pi r_H^2. \quad (15)$$

The entropy is therefore

$$S_{BH} = \eta k_B N_H = \eta k_B \frac{A_H}{\ell_*^2}, \quad (16)$$

with dimensionless counting coefficient η left temporarily unfixed. Since

$$A_H = 16\pi G_{\text{eff}}^2 M_{\text{inv}}^2 / c^4,$$

we have

$$\frac{dS_{BH}}{dM_{\text{inv}}} = 32\pi\eta k_B \frac{G_{\text{eff}}^2 M_{\text{inv}}}{c^4 \ell_*^2}. \quad (17)$$

The coefficient η and the calibration of ℓ_* will be fixed by consistency with the temperature derived in Section 7.2 and the first-law closure of Section 7.3. Until then, (16) is the generic boundary-count entropy law implied by CF03 plus the saturation geometry of Section 6.

7.2 7.2 Hawking temperature from re-articulation rate

At the horizon the same-domain channel saturates. CF000 therefore forces orthogonal re-articulation. The semiclassical exterior reading of that re-articulation is thermal emission. The characteristic energy scale is controlled by the surface gravity. For the static exterior metric (13), define

$$f(r) := 1 - \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2 r}, \quad g_{tt} = -c^2 f(r).$$

Then

$$\kappa_H = \frac{c^2}{2} |f'(r_H)| = \frac{1}{2} |\partial_r g_{tt}|_{r_H}$$

with

$$f'(r) = \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2 r^2}, \quad r_H = \frac{2G_{\text{eff}}M_{\text{inv}}}{c^2}.$$

Substituting r_H gives

$$\kappa_H = \frac{c^4}{4G_{\text{eff}}M_{\text{inv}}}. \quad (18)$$

The associated temperature is

$$T_H = \frac{\hbar \kappa_H}{2\pi k_B c} = \frac{\hbar c^3}{8\pi k_B G_{\text{eff}} M_{\text{inv}}}. \quad (19)$$

Equation (19) is the completed Hawking-temperature law of CF12. Its meaning in the present canon is direct: the steeper the local saturation geometry, the higher the rate at which orthogonal re-articulation is forced into the outgoing channel.

Now fix the boundary-count coefficient by first-law consistency. Demanding $d(M_{\text{inv}} c^2) = T_H dS_{BH}$ with (17) and (19) yields

$$\ell_*^2 = 4\eta \frac{\hbar G_{\text{eff}}}{c^3}.$$

Choosing the natural gravitational articulation scale

$$\ell_*^2 := \frac{\hbar G_{\text{eff}}}{c^3} \quad (20)$$

therefore fixes

$$\eta = \frac{1}{4}. \quad (21)$$

Hence the completed horizon entropy law is

$$S_{BH} = \frac{k_B A_H}{4\ell_*^2} = \frac{k_B c^3 A_H}{4\hbar G_{\text{eff}}}. \quad (22)$$

This is the Bekenstein–Hawking area law as the boundary articulation count of the saturation surface.

7.3 First law and evaporation rate

Using (22) and (19), the thermodynamic first law closes as

$$dE = c^2 dM_{\text{inv}} = T_H dS_{BH}. \quad (23)$$

Approximating the outgoing channel as effective blackbody emission with horizon area A_H , one gets the leading quasi-static mass-loss law

$$\frac{dM_{\text{inv}}}{dt} = -\frac{\gamma \hbar c^4}{G_{\text{eff}}^2 M_{\text{inv}}^2}, \quad (24)$$

where γ is a dimensionless greybody-corrected emission coefficient. CF12 does not need a primitive emission postulate; it needs only the saturation temperature and the existence of the outgoing re-articulation channel.

8. Information Ledger and the Page Curve

8.1 Information as articulation record

Definition 8.1 (Articulation record). The information carried by a channel is its invariant articulation record, denoted $\mathcal{I}$. For a black-hole system split into horizon/interior channel H , radiation channel R , and asymptotic exterior environment E , the total record is

$$\mathcal{I}_{\text{tot}} = \mathcal{I}_H + \mathcal{I}_R + \mathcal{I}_E. \quad (25)$$

Because CF000 forbids discharge, $\mathcal{I}_{\text{tot}}$ is preserved by redistribution even when any one channel saturates.

Theorem 8.2 (Information preservation under evaporation). *Under horizon evaporation,*

$$\frac{d\mathcal{I}_{\text{tot}}}{dt} = 0, \quad \frac{d\mathcal{I}_H}{dt} < 0, \quad \frac{d\mathcal{I}_R}{dt} > 0, \quad (26)$$

with $d\mathcal{I}_R/dt = -d\mathcal{I}_H/dt$ after subtracting any asymptotic leakage into the environment channel.

Proof. The horizon channel is saturated and therefore cannot continue as an isolated same-domain host. CF000 demands orthogonal re-articulation. The outgoing radiation channel is exactly that new host. The record therefore transfers channel, rather than disappearing. The total articulation record is preserved because non-discharge blocks its destruction. $\square$

8.2 Page curve as transfer law

Let

$$Q(t) := \int_0^t \frac{1}{T_H(t')} \left(-\frac{dE}{dt'} \right) dt' = \int_0^t \dot{S}_{\text{emit}}(t') dt' \quad (27)$$

be the cumulative coarse-grained emitted entropy. The coarse-grained radiation entropy is then

$$S_R(t) = \min\{Q(t), S_{BH}(0) - Q(t)\}, \quad (28)$$

with Page time t_P defined by

$$Q(t_P) = \frac{1}{2} S_{BH}(0). \quad (29)$$

Equation (28) is the completed Page-curve law of CF12. Its interpretation is immediate: early radiation grows in coarse-grained entropy while the outgoing channel accumulates inaccessible record; once roughly half the initial horizon record has transferred, further transfer purifies the radiation channel rather than entropizing it.

9. Cosmogenesis Residual Closure by Gravitational Back-Reaction

CF11 leaves a residual that this module is allowed to address. The completed CF12 correction is not vague: it is an explicit gravitational back-reaction sourced by emergent potential wells and saturation gradients generated on top of the CF11 background.

Define the modified cosmological closure by

$$H^2(a) = H_{11}^2(a) + \xi \langle |\nabla \Phi|^2 \rangle_a + \zeta \langle \sigma_{\text{sat}} \rangle_a, \quad (30)$$

where $H_{11}(a)$ is the background Hubble law inherited from CF11, Φ solves the weak-field Poisson equation (8) on the coarse-grained cosmological carrier, and σ_{sat} is the carrier-average saturation-production density associated with $s(x)$. The angle brackets denote the cosmological coarse-graining at scale factor a . The parameters ξ and ζ are not arbitrary fitting decorations. They are the effective couplings of two already-earned gravitational sectors: potential-well seeding and saturation back-reaction.

Remark 9.1. Equation (30) is the completed CF12 residual-closure ansatz. It is explicit enough to be tested directly against the CF11 residual. If the best-fit (ξ, ζ) fail to reduce the residual or degrade the previously matched observables, the correction sector must be rejected or refined. But the formal law itself is not missing.

10. Validation Gates

Because CF12 is now written as a completed formalism, its decisive tests are equally explicit.

10.1 Weak-field gates

- Poisson recovery: the 00-sector of (6) must reduce to (8).
- Schwarzschild recovery: static spherical vacuum must reproduce (13).
- Redshift sign and scaling: the metric coefficient g_{00} must give standard attractive redshift in the weak field.

10.2 Horizon and thermodynamic gates

- Horizon condition: $\chi(r_H) = 1$ must coincide with the null surface of the metric.
- Area law: entropy must scale with A_H rather than volume.
- Temperature law: emission temperature must obey (19) to leading semiclassical order.
- First law: (23) must hold under quasi-static evaporation.

10.3 Information and cosmology gates

- Page curve: numerical transfer models based on (28) must exhibit rise, peak, and fall.
- Information preservation: no evaporation model is admissible if it destroys $\mathcal{I}_{\text{tot}}$.
- Cosmogenesis residual: the correction law (30) must reduce the CF11 residual without degrading previously matched background observables.

11 11. Red-Team Leak Checklist

- R1. Primitive metric leak.** Any use of a metric before the cone-to-geometry derivation is illegal.
- R2. Signature familiarity leak.** Lorentzian signature is admissible only because the inherited cone has one common propagation-limiting direction.
- R3. Einstein familiarity leak.** Einstein form is admissible only as the minimal local second-order divergence-compatible closure of burden-gradient geometry.
- R4. Horizon-annihilation leak.** Any reading of the horizon as information destruction contradicts CF000.
- R5. Entropy-as-bulk leak.** Any attempt to justify the area law without CF03 boundary concentration is a stage leak.
- R6. Residual-handwave leak.** The cosmogenesis correction must be explicit, not gestural.

12 12. Completed Formalism Statement

The completed CF12 formalism is the following system.

1. Carrier source bookkeeping:

$$T_{\mu\nu}^{\text{inv}} = \rho_{\text{inv}} u_\mu u_\nu + p_{\text{inv}} P_{\mu\nu} + \Pi_{\mu\nu} + q_{(\mu} u_{\nu)}.$$

2. Equivalence principle:

$$m_I = m_G = c^{-2} \mathcal{B}.$$

3. Lorentzian cone metric:

$$ds^2 = -c_*^2 d\tau^2 + h_{ij} dx^i dx^j, \quad g^{\mu\nu} \partial_\mu S \partial_\nu S = 0.$$

4. Effective gravitational field equation:

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \Delta_{\mu\nu}^{\text{sat}} = \kappa_{\text{inv}} T_{\mu\nu}^{\text{inv}}.$$

5. Weak-field closure:

$$\nabla^2 \Phi = 4\pi G_{\text{eff}} \rho_{\text{inv}}.$$

6. Static spherical exterior:

$$ds^2 = - \left(1 - \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r} \right) c^2 dt^2 + \left(1 - \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2 r} \right)^{-1} dr^2 + r^2 d\Omega^2.$$

7. Horizon and area law:

$$r_H = \frac{2G_{\text{eff}} M_{\text{inv}}}{c^2}, \quad S_{BH} = \frac{k_B A_H}{4\ell_*^2}, \quad \ell_*^2 = \frac{\hbar G_{\text{eff}}}{c^3}.$$

8. Hawking temperature and first law:

$$T_H = \frac{\hbar c^3}{8\pi k_B G_{\text{eff}} M_{\text{inv}}}, \quad d(M_{\text{inv}} c^2) = T_H dS_{BH}.$$

9. Page-curve law:

$$S_R(t) = \min\{Q(t), S_{BH}(0) - Q(t)\}, \quad Q(t) = \int_0^t \dot{S}_{\text{emit}} dt'.$$

10. Cosmogenesis residual closure:

$$H^2(a) = H_{11}^2(a) + \xi \langle |\nabla \Phi|^2 \rangle_a + \zeta \langle \sigma_{\text{sat}} \rangle_a.$$

These ten lines are the compact completed gravity module of the present canon. The preceding sections derive their placement, interpretation, and mutual compatibility.

A Appendix A. Inherited Result Ledger

Source	Inherited result	Use in CF12
CF000	Non-discharge, saturation, or- thogonal re-articulation	Horizon as saturation; radiation as re-articulation; in- formation preservation
CF00	Carrier $\mathcal{M}$, smooth local varia- tion	Host on which burden density and geometry are writ- ten
CF01	$J \oplus M$ split, scalar generators I and Σ	Reversible/irreversible reading of transport, source bookkeeping, and saturation
CF03	Boundary concentration under hi- erarchical saturation	Area law and boundary-hosted entropy
CF04	Finite-speed cone	One timelike direction and Lorentzian signature
CF11	Dark-sector cosmological back- ground	Baseline for residual closure term

B Appendix B. Minimal Symbol Table

Symbol	Meaning
$\mathcal{M}$	Derived carrier domain
ρ_{inv}	Invariant burden density
$\mathcal{B}$	Total invariant burden functional on a host region
u^μ	Carrier four-velocity field
$P_{\mu\nu}$	Spatial projector orthogonal to u^μ
$T_{\mu\nu}^{\text{inv}}$	Effective burden source tensor
s	Saturation scalar
$\Delta_{\mu\nu}^{\text{sat}}$	Saturation correction tensor
G_{eff}	Effective gravitational coupling
Λ_{eff}	Effective cosmological background term
Φ	Weak-field gravitational potential
M_{inv}	Total mass-equivalent burden of a compact source
r_H	Horizon radius
A_H	Horizon area
ℓ_*	Fundamental boundary articulation length scale
S_{BH}	Black-hole entropy
T_H	Hawking temperature
$\mathcal{I}$	Articulation record / information content
$Q(t)$	Cumulative emitted coarse-grained entropy